

Points	Criteria
30	Correctly implements realtime ray tracing using compute shaders and WebGPU. Performs decently well - more than 30 FPS on average.
20	Add Interactivity with a GUI where you can switch between different scenes to illustrate different ray tracing concepts, adjust light parameters, and adjust material parameters. Allow visualization of the Bounding Volume Hierarchy.
10	Correctly implement reflection, refraction, and diffuse material types for your objects. Make the scene look smooth with minimal noise. Have colors look realistic by applying gamma correction.
10	Create complex scenes with many objects with each object being customizable in terms of their position, scale, color, and material properties. Allow the user to move around the scene and look in any direction.
70	Total

Points earned:

+30 - I implemented ray tracing using compute shader and the default scene has a lot more than 30fps on average

+20 - I added a GUI where I can switch between different scenes and have BVH visualization.

+10 - implemented reflection, refraction, diffuse materials, added accumulation buffer to reduce noise, added gamma correction.

+8 - Added a second scene which can have a variable number of spheres, can move in the scene. Could have added more object types.